OPMIG-05: Dynamic Routing & Bucket Management

> **Series:** OPMIG — OpenPipeline Migration | **Notebook:** 5 of 10 | **Created:** December 2025 | **Last Updated:** 10/26/2026

Table of Contents

1. Understanding Dynamic Routing
2. Routing Strategies
3. Grail Buckets Explained
4. Bucket Routing Configuration
5. Cost Optimization Patterns
6. Cost Optimization ROI Calculator ⭐ NEW
7. Analyzing Your Bucket Usage
8. Multi-Tier Bucket Strategies ⭐ NEW
9. Multi-Pipeline Processing
10. Bucket Governance & Access Control ⭐ NEW
11. Best Practices
12. Routing Configuration Examples

Learning Objectives

By completing this notebook, you will:

1. Master dynamic routing strategies for different migration scenarios
2. Design multi-tier bucket architectures for cost optimization

3. ⭐ **NEW:** Calculate ROI and TCO for bucket optimization strategies
4. ⭐ **NEW:** Implement 3-tier and 5-tier bucket governance models
5. ⭐ **NEW:** Apply compliance-driven retention strategies
6. Configure multi-pipeline processing patterns
7. Analyze bucket usage and identify optimization opportunities

---

Prerequisites

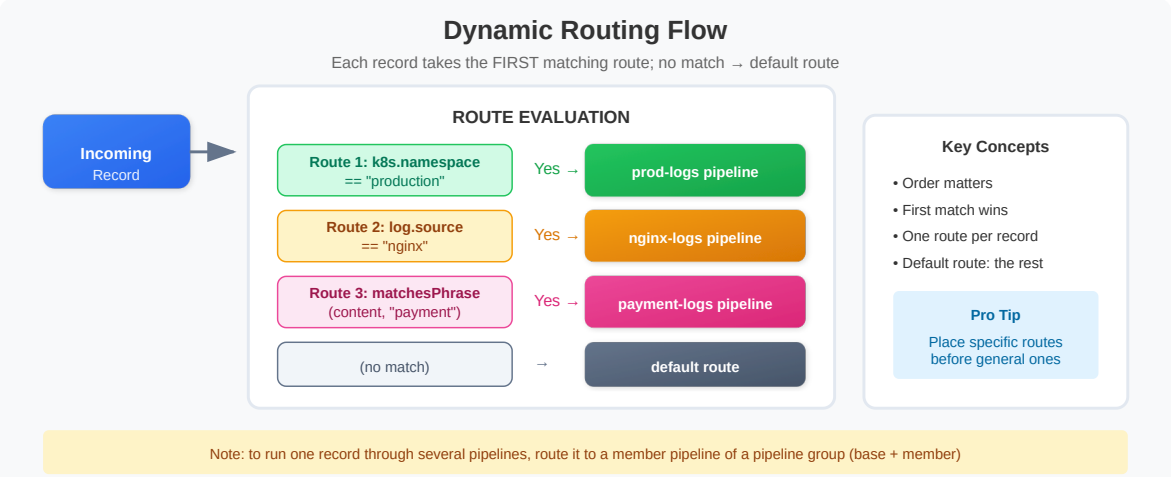
Requirement	Details
Dynatrace Environment	Dynatrace SaaS with Grail and OpenPipeline access — Managed is not covered by this series
Permissions	`settings:read`, `settings:write` (OpenPipeline configuration), `storage:buckets:read`
API Access	`logs.read` token scope
Knowledge	OPMIG-01 through OPMIG-04; understanding of pipeline configuration

---

Understanding Dynamic Routing

Dynamic routing is the mechanism that directs incoming data to specific pipelines based on matching conditions.

How Routing Works



Key Routing Concepts

Concept	Description
Matching Condition	DQL expression that evaluates to true/false
Route Priority	Routes are evaluated in order (top to bottom) — the **first matching route wins**
One route per record	A record that matches several routes still goes to only the first one
Default route	Catches all records not matching any route — for logs and business events, the classic pipeline
More than one pipeline	Route to a member pipeline of a **pipeline group**; the group's base pipelines run with it

> **Sources:** Data flow in OpenPipeline (DT docs) — *"The route order is relevant—the position in the list establishes the order of execution."*; Pipeline groups — multi-cloud ingest governance (DT docs) — *"Each record is routed to and processed by only one member pipeline, according to the first matching route."*

Routing Strategies

Strategy 1: Source-Based Routing

Route based on data source or ingestion method:

Condition	Target Pipeline	Use Case
`log.source == "nginx"`	nginx-logs	Web server logs
`log.source == "application"`	app-logs	Application logs
`dt.openpipeline.source == "/api/v2/otlp/v1/logs"`	otel-logs	OpenTelemetry (OTLP logs)
`dt.openpipeline.source == "/api/v2/logs/ingest"`	api-logs	API ingestion

> For built-in API sources, `dt.openpipeline.source` holds the endpoint **path** (`oneagent` for OneAgent). Run OPMIG-02's *View data sources* query to see the values in your tenant before routing on this field.

Strategy 2: Environment-Based Routing

Route based on environment or namespace:

Condition	Target Pipeline	Bucket
`k8s.namespace.name == "production"`	prod-logs	default_logs (35d)
`k8s.namespace.name == "staging"`	staging-logs	dev_logs (7d)
`k8s.namespace.name == "development"`	dev-logs	dev_logs (7d)

Strategy 3: Content-Based Routing

Route based on log content patterns:

Condition	Target Pipeline	Purpose
`matchesPhrase(content, "payment")`	payment-logs	Financial logs
`matchesPhrase(content, "security")`	security-logs	Security events
`matchesPhrase(content, "audit")`	audit-logs	Compliance logs

> Routing matchers accept `matchesPhrase` (whole tokens or phrases) and `matchesValue` with `*` wildcards (any substring); `contains()` is not enabled in matchers.

Strategy 4: Severity-Based Routing

Route based on log level for tiered retention:

Condition	Target Pipeline	Bucket
`status == "ERROR"`	error-logs	error_logs (90d)
`status == "WARN"`	warning-logs	default_logs (35d)
`loglevel == "DEBUG"`	debug-logs	short_retention (3d)

> Use `status` for error- and warning-class tiers. `status` groups the levels — `SEVERE`, `ERROR`, `CRITICAL`, `ALERT`, `FATAL` and `EMERGENCY` all become `ERROR` — so `loglevel == "ERROR"` alone misses Java `SEVERE` and the rest (on a validation tenant, 63,216 `loglevel == "ERROR"` records against 118,023 `status == "ERROR"` in one hour, 10/14/2026). `status` folds `DEBUG` and `TRACE` into `INFO`, so keep `loglevel` for the debug tier.
>
> **Sources:** Automatic log enrichment (DT docs) — *"Additionally, for each log event, a status attribute is created with a value that is a sum of loglevel values based on the following grouping".*

Strategy 5: Hybrid Routing

Combine multiple criteria for precise routing:

```
k8s.namespace.name == "production" AND log.source == "payment-service"
```

Grail Buckets Explained

Grail buckets are storage containers with configurable retention and access policies.

Default Buckets

Bucket	Data Type	Default Retention
`default_logs`	Log records	35 days
`default_spans`	Span/trace data	10 days
`default_bizevents`	Business events	35 days
`default_events`	Platform events	35 days

Run `fetch dt.system.buckets | filter startsWith(name, "default_")` to see the built-in buckets and retention in your own tenant.

> **Sources:** Organize data (DT docs) — built-in bucket table: `default_logs` 35 days, `default_spans` 10 days, `default_bizevents` 35 days, `default_events` 35 days.

Custom Bucket Use Cases

Use Case	Bucket Strategy	Retention
Compliance/Audit	`audit_logs`	365+ days
Financial Data	`financial_logs`	90 days
Security Events	`security_logs`	180 days
Development Logs	`dev_logs`	7 days
Debug/Trace	`debug_logs`	3 days
High-Volume Noise	`ephemeral_logs`	1 day

Bucket Benefits

1. **Cost Optimization**: Shorter retention = lower storage costs
2. **Compliance**: Longer retention for audit requirements
3. **Performance**: Smaller buckets = faster queries
4. **Access Control**: Bucket-level permissions
5. **Data Isolation**: Separate buckets for different teams/purposes

---

Bucket Routing Configuration

Configuring Bucket Routing in Pipeline

Each pipeline can route data to a specific bucket:

1. Open your pipeline in OpenPipeline settings
2. Go to the **Bucket assignment** stage and add a **Bucket assignment** processor
3. Give it a matching condition and choose the bucket from the **Storage** list
4. Save pipeline

Bucket Routing Rules

Rule	Behavior
First match only	The first Bucket assignment processor whose condition matches decides the bucket
No storage assignment	The record runs through every later stage (metric, Davis, data extraction) and is then not stored
No processor matched	In community practice the record lands in the scope's default bucket (`default_logs`) — confirm with the bucket queries below

> ⚠️ **Important:** A record is stored at most **once**, in one bucket. Bucket assignment is first-match-only, so order the processors from most specific to least specific.

Creating Custom Buckets

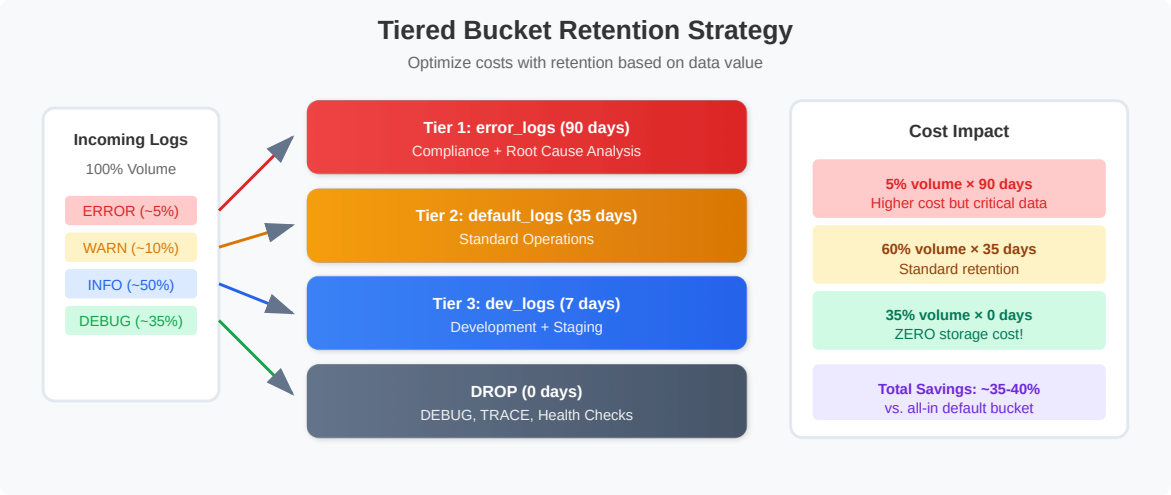
Custom buckets are created via Grail settings:

```
Settings → Storage management → Bucket storage management → Bucket
```

Configure:
- **Name**: Descriptive bucket name
- **Table**: Data type (logs, spans, etc.)
- **Retention**: Days to retain data
- **Description**: Purpose of bucket

Cost Optimization Patterns

Pattern 1: Tiered Retention by Severity



Pattern 2: Environment-Based Retention

Environment	Bucket	Retention	Cost Impact
Production	`prod_logs`	35 days	Standard
Staging	`staging_logs`	14 days	-60%
Development	`dev_logs`	7 days	-80%

Environment	Bucket	Retention	Cost Impact
Ephemeral/Test	`temp_logs`	3 days	-90%

Pattern 3: Drop + Sample

For extremely high-volume, low-value logs:

1. Drop 90% of health checks
2. Sample 10% to a short-retention bucket
3. Keep statistical visibility without full storage cost

Pattern 4: Metric Extraction + No Storage

For logs needed only for metrics:

1. Extract metrics (with dimensions) from logs in the **Metric extraction** stage
2. Assign the raw logs **No storage assignment** (Bucket assignment stage) so each record still runs through Metric extraction but is not stored
3. The metrics are kept for the metrics bucket's retention (`default_metrics` : 15 months); the raw logs are never written to a bucket

Do not use a **Drop record** processor for this — it runs in the Processing stage, before Metric extraction, so nothing would be extracted.

> **Sources:** Processing in OpenPipeline (DT docs) — Drop record: *"Drops a record. The record isn't processed further and isn't stored."*; No storage assignment: *"The record continues through all configured pipeline stages and isn't stored only at the end of the pipeline."*

Cost Optimization ROI Calculator ⭐ NEW

Understanding the financial impact of bucket strategies is critical for migration planning. This section shows how to estimate the effect of a bucket strategy — the **arithmetic**, not a price list.

> ⚠️ **Placeholder rates — not Dynatrace prices.** Every rate below (the per-MB DDU rates and the $0.08 per DDU) is an illustrative input chosen to show the arithmetic. No Dynatrace page publishes these figures. Replace each one with the rates from your own contract before using any result. On a **Dynatrace Platform Subscription (DPS)**, log costs are billed per capability (Ingest & Process, Retain, Query) — see FINOPS-01 and your rate card. The relative result — storage cost scales with volume × retention days — holds whatever the rates are; the query below (*Estimate storage reduction from tiered retention*) computes it from your own data.

Pricing Model (placeholder inputs)

> **Note:** DPS is the **Dynatrace Platform Subscription**. DDUs (**Davis data units**) are the classic-licensing unit for custom metrics, Log Monitoring and custom events. Check your license model to determine which applies.

Placeholder inputs used in the scenarios below:

Component	Placeholder rate
Log Ingestion	0.001 DDU per MB ingested
Storage (per day)	0.0001 DDU per MB per day retained
DQL Query	0.001 DDU per MB scanned

Placeholder DDU price: $0.08 per DDU

Cost Formula

```
Monthly Cost = (Ingestion Cost) + (Storage Cost) + (Query Cost)

Where:
  Ingestion Cost = Volume (MB/day) × 30 days × Ingest rate × Price/DDU
  Storage Cost   = Volume (MB/day) × Retention (days) × Storage rate (per MB
per day) × 30 days × Price/DDU
                   (steady state: Volume × Retention MB are held at any time,
charged every day)
  Query Cost     = Query Volume (MB/month) × Query rate × Price/DDU
```

Scenario 1: E-Commerce Platform - Tiered Retention

Current State (No Optimization):

```
Volume:          100 GB/day (100,000 MB/day)
Breakdown:       - ERROR:  5% (5,000 MB/day)
                 - WARN:  10% (10,000 MB/day)
                 - INFO:  50% (50,000 MB/day)
                 - DEBUG: 35% (35,000 MB/day)
Retention:       35 days (default_logs)
DDU Price:       $0.08 per DDU (placeholder)
```

Current Monthly Cost:

```
Ingestion:  100,000 MB/day × 30 days × 0.001 × $0.08 = $240/month
Storage:    100,000 MB/day × 35 days × 0.0001 × 30 days × $0.08 = $840/month
Query:      Estimate ~20% of monthly volume queried
            (100,000 × 30 × 0.2) × 0.001 × $0.08 = $48/month

TOTAL:      $1,128/month
```

Optimized State (3-Tier + Drop):

```
Tier 1 (ERROR):     5,000 MB/day → error_logs (90 days)
Tier 2 (WARN/INFO): 60,000 MB/day → default_logs (35 days)
Tier 3 (DEBUG):     35,000 MB/day → DROPPED (0 days)
```

Optimized Monthly Cost:

```
Ingestion (only non-dropped — assumes dropped records carry no ingest charge;
check your rate card):
  65,000 MB/day × 30 days × 0.001 × $0.08 = $156/month

Storage:
  Tier 1: 5,000 × 90 × 0.0001 × 30 × $0.08 = $108/month
  Tier 2: 60,000 × 35 × 0.0001 × 30 × $0.08 = $504/month
  Total storage = $612/month

Query (20% of monthly volume):
  (5,000 × 30 × 0.2 + 60,000 × 30 × 0.2) × 0.001 × $0.08 = $31.20/month

TOTAL:      $799.20/month
```

ROI Analysis:

```
Monthly Savings:     $1,128 - $799.20 = $328.80/month
Annual Savings:      $328.80 × 12 = $3,945.60/year
Savings Percentage:  29.1%
Implementation Time: 4-8 hours
Payback Period:      Immediate (operational change only)
```

Scenario 2: SaaS Platform - Multi-Environment

Storage cost only, same placeholder rates:

Current State:

```
Production:   50 GB/day × 35 days = $420/month
Staging:      30 GB/day × 35 days = $252/month
Development:  20 GB/day × 35 days = $168/month
TOTAL:        $840/month
```

Optimized State:

```
Production:   50 GB/day × 35 days = $420/month (no change)
Staging:      30 GB/day × 14 days = $100.80/month (60% reduction)
Development:  20 GB/day × 7 days  = $33.60/month (80% reduction)
TOTAL:        $554.40/month
```

ROI Analysis:

```
Monthly Savings:     $285.60/month
Annual Savings:      $3,427.20/year
Savings Percentage:  34%
```

Scenario 3: Global Retailer - Compliance + Cost

Requirements:
- Payment logs: PCI DSS Requirement 10.5.1 — at least 12 months of audit log history, the most recent three months immediately available
- Audit logs: 365 days — SOC 2 sets no fixed period; a year matching the audit window is common practice
- Application logs: 35 days standard
- Debug logs: Drop immediately

Volume Breakdown:

```
Payment logs:  10 GB/day (10% of total)
Audit logs:    5 GB/day  (5% of total)
App logs:      60 GB/day (60% of total)
Debug logs:    25 GB/day (25% of total)
TOTAL:         100 GB/day
```

Cost Calculation:

```
Payment (365d): 10,000 × 365 × 0.0001 × 30 × $0.08 = $876/month
Audit (365d):   5,000 × 365 × 0.0001 × 30 × $0.08 = $438/month
App (35d):      60,000 × 35 × 0.0001 × 30 × $0.08 = $504/month
Debug (drop):   $0/month
Ingestion:      75,000 × 30 × 0.001 × $0.08 = $180/month

TOTAL:          $1,998/month
```

vs. No Optimization (all 35 days):

```
Storage:   100,000 × 35 × 0.0001 × 30 × $0.08 = $840/month
Ingestion: 100,000 × 30 × 0.001 × $0.08       = $240/month
TOTAL:     $1,080/month (but non-compliant!)
```

Key Insight: Sometimes optimization increases cost but ensures compliance. The cost of non-compliance (fines, audits) far exceeds storage costs.

> **Sources:** Record deletion in Grail via API (DT docs) — DPS expansion, *"This must be enabled as a capability in your Dynatrace Platform Subscription (DPS)."*; Davis data units (DT docs) — *"Davis data units (DDU) provide a simple means of licensing certain capabilities (custom metrics, log monitoring, and custom events) on the Dynatrace platform."*

Analyzing Your Bucket Usage

Use these queries to understand your current bucket utilization and plan optimization.

```
// Current bucket distribution for logs
// Shows where your logs are being stored
fetch logs, from: now() - 7d
| summarize {record_count = count()}, by: {dt.system.bucket}
| fieldsAdd daily_avg = round(record_count / 7, decimals: 0)
| sort record_count desc
```

```
// Bucket usage by log source
// Identify which sources are consuming bucket capacity
fetch logs, from: now() - 7d
| summarize {record_count = count()}, by: {dt.system.bucket, log.source}
| sort record_count desc
| limit 30
```

```
// Bucket usage by pipeline
// Shows which pipelines are routing to which buckets
fetch logs, from: now() - 7d
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {record_count = count()}, by: {dt.openpipeline.pipelines,
dt.system.bucket}
| sort record_count desc
```

```
// Bucket usage trend over time
fetch logs, from: now() - 7d
| makeTimeseries {record_count = count()}, by: {dt.system.bucket}, interval:
24h
```

```
// Identify logs that could move to shorter retention buckets
// Debug and trace logs are prime candidates
fetch logs, from: now() - 7d
| filter dt.system.bucket == "default_logs"
| summarize {
    total = count(),
    debug = countIf(loglevel == "DEBUG" OR status == "DEBUG"),
    trace = countIf(loglevel == "TRACE" OR status == "TRACE"),
    info = countIf(loglevel == "INFO" OR status == "INFO")
  }
| fieldsAdd debug_pct = round((toDouble(debug) / toDouble(total)) * 100,
decimals: 2)
| fieldsAdd trace_pct = round((toDouble(trace) / toDouble(total)) * 100,
decimals: 2)
| fieldsAdd optimization_potential = round((toDouble(debug + trace) /
toDouble(total)) * 100, decimals: 2)
```

```
// Check span bucket distribution
// Spans also use buckets
fetch spans, from: now() - 7d
| summarize {span_count = count()}, by: {dt.system.bucket}
| sort span_count desc
```

```
// Estimate storage reduction from tiered retention
// Shows potential savings from routing to different buckets
// Error/warn tier uses status, which also counts SEVERE, CRITICAL, FATAL and
the rest
fetch logs, from: now() - 7d
| summarize {
    total = count(),
    error_warn = countIf(status == "ERROR" OR status == "WARN"),
    info = countIf(loglevel == "INFO"),
    debug_trace = countIf(loglevel == "DEBUG" OR loglevel == "TRACE")
  }
| fieldsAdd tier1_90day = error_warn
| fieldsAdd tier2_35day = info
| fieldsAdd tier3_drop = debug_trace
| fieldsAdd current_cost_units = total * 35
| fieldsAdd optimized_cost_units = (tier1_90day * 90) + (tier2_35day * 35) +
(tier3_drop * 0)
| fieldsAdd savings_pct = round((1.0 - (toDouble(optimized_cost_units) /
toDouble(current_cost_units))) * 100, decimals: 1)
```

Multi-Tier Bucket Strategies ⭐ NEW

Different organizations need different bucket architectures based on their scale, compliance requirements, and cost targets.

3-Tier Strategy (Small to Medium Organizations)

Recommended for: 10-500 GB/day, single or few environments

Tier	Bucket Name	Retention	Use Case	Volume %
Critical	`critical_logs`	90 days	Errors, security, audit	10-15%
Standard	`default_logs`	35 days	WARN, INFO, app logs	60-70%

Tier	Bucket Name	Retention	Use Case	Volume %
Ephemeral	`ephemeral_logs`	7 days	DEBUG, health checks	20-30%

Routing Configuration:

```
Route 1 (Critical):
  Condition: status == "ERROR" OR
             matchesValue(log.source, "*security*") OR
             matchesValue(log.source, "*audit*") OR
             matchesPhrase(content, "payment")
  Pipeline:  critical-logs
  Bucket:    critical_logs (90 days)

Route 2 (Ephemeral):
  Condition: loglevel == "DEBUG" OR
             loglevel == "TRACE" OR
             matchesValue(content, "*/health*")
  Pipeline:  ephemeral-logs
  Bucket:    ephemeral_logs (7 days)

Route 3 (Standard, catch-all — last in the list):
  Condition: true
  Pipeline:  standard-logs
  Bucket:    default_logs (35 days)
```

Expected savings: depends on your level mix — run *Estimate storage reduction from tiered retention* above on your own data before committing to a number.

5-Tier Strategy (Enterprise Organizations)

Recommended for: 500+ GB/day, multiple environments, strict compliance

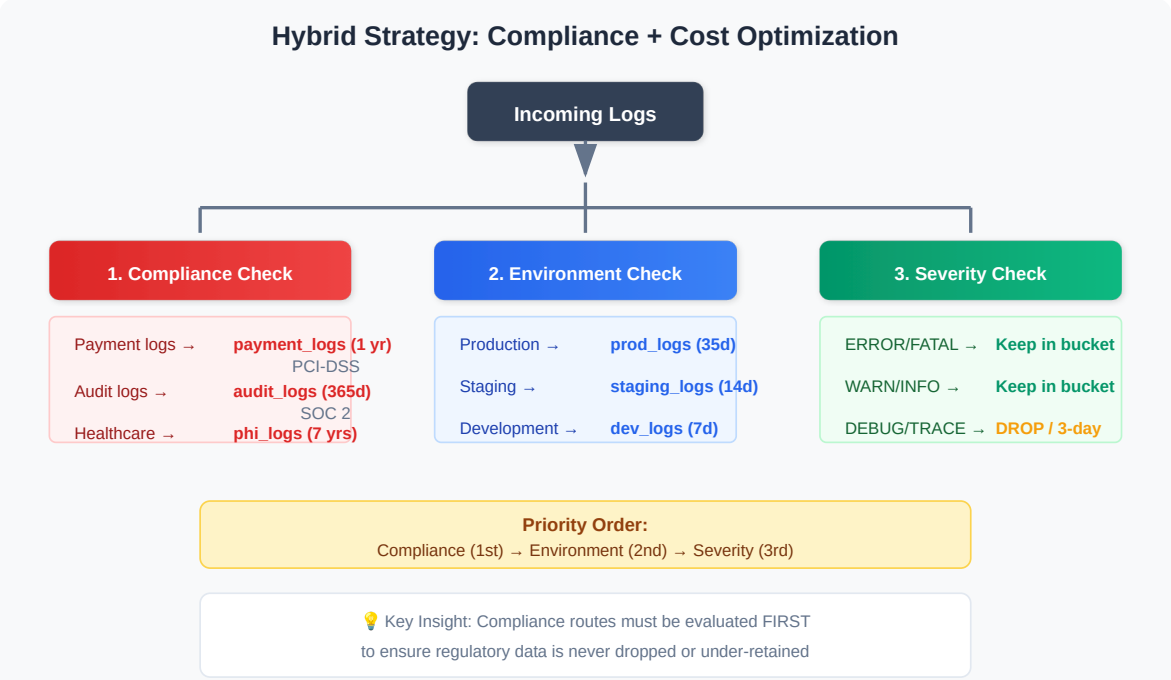
Tier	Bucket Name	Retention	Use Case	Volume %
Compliance	`compliance_logs`	365 days	SOC 2, GDPR, HIPAA audit trails	3-5%
Security	`security_logs`	180 days	Security events, attacks, vulns	2-5%
Production	`prod_logs`	90 days	Production errors, warnings	10-15%

Tier	Bucket Name	Retention	Use Case	Volume %
Standard	`default_logs`	35 days	INFO logs, standard operations	40-50%
Development	`dev_logs`	7 days	Dev/staging, DEBUG, testing	30-40%

Expected savings: measure, as above — the volume percentages in the table are illustrative.

Hybrid Strategy (Compliance + Cost)

Best for: Organizations with compliance requirements AND cost constraints

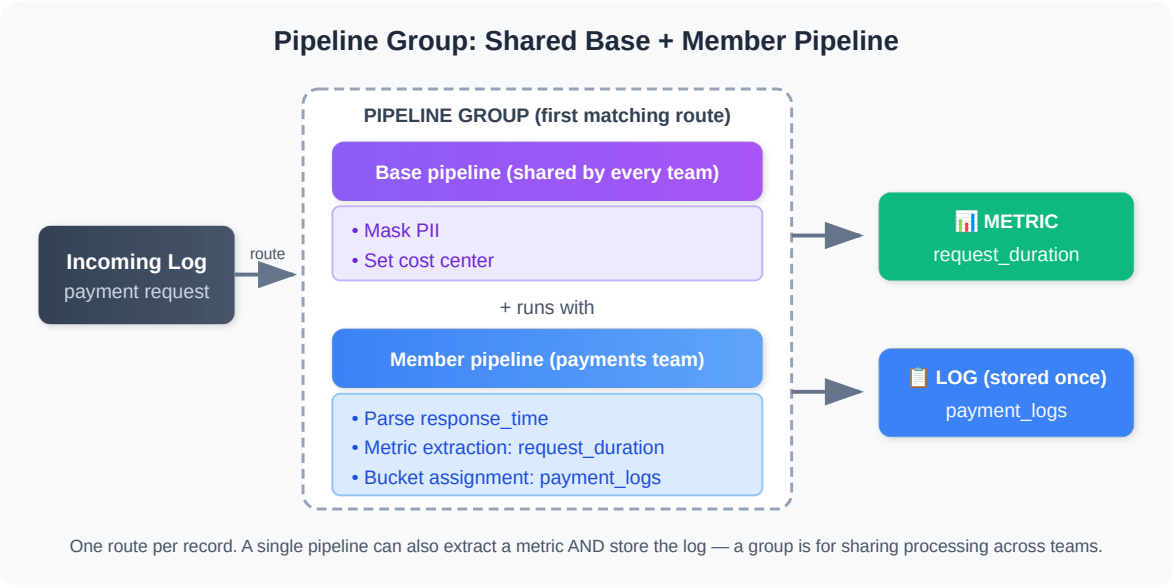


Key Principle: Compliance routes must be FIRST to ensure regulatory data isn't accidentally dropped or under-retained.

Multi-Pipeline Processing

Routing sends each record to **one** route — the first that matches. To run a record through more than one pipeline, use a **pipeline group**: a route targets a *member* pipeline, and the group's *base* pipelines (shared processing such as masking, cost and permission assignment) run with it.

Use Case: Shared Base + Team Member Pipeline



You do **not** need two pipelines to extract a metric and store the log — one pipeline does both (Metric extraction and Bucket assignment are stages of the same pipeline). Use a group when several teams need the same baseline processing and their own additions on top.

Configuring a Pipeline Group

1. Create the shared processing as a **base pipeline** and the team-specific processing as a **member pipeline**, both in one pipeline group
2. Route the data to the **member pipeline** with a dynamic route
3. The record is processed by the base pipelines and that member pipeline — overlapping *routes* do not do this; the second route never sees a record the first one matched

Base pipelines are not routable — *"base pipelines are typically not routable"* — so every route targets a member. Only administrators assign members to a group: *"Only*

administrators can manage member pipeline assignment to a group." For ownership, stage allocation, execution order and gating, see **OPIPE-07: Pipeline Group Governance Patterns**.

> ⚠️ **Limit:** Data extraction happens in at most 5 pipelines for one record (`dt.openpipeline.pipelines`). Beyond that, extraction stops; the record is still processed and persisted.
>
> **Sources:** OpenPipeline pipeline groups (DT docs), Pipeline groups — multi-cloud ingest governance (DT docs) — *"A record is processed by base pipelines and a member pipeline, according to the first matching route."*; OpenPipeline limits (DT docs) — *"You can extract data on a single record in a maximum of five different pipelines"*.

```
// Identify records processed by multiple pipelines
fetch logs, from: now() - 1h
| filter isNotNull(dt.openpipeline.pipelines)
| filter arraySize(dt.openpipeline.pipelines) &gt; 1
| summarize {multi_pipeline_count = count()}, by: {dt.openpipeline.pipelines}
| sort multi_pipeline_count desc
| limit 20
```

```
// Count of logs by number of pipelines processing them
fetch logs, from: now() - 1h
| filter isNotNull(dt.openpipeline.pipelines)
| fieldsAdd pipeline_count = arraySize(dt.openpipeline.pipelines)
| summarize {record_count = count()}, by: {pipeline_count}
| sort pipeline_count asc
```

Bucket Governance & Access Control ⭐ NEW

Proper bucket governance ensures compliance, security, and operational efficiency.

Access Control Patterns

Pattern 1: Role-Based Bucket Access

People read log buckets; only the pipeline writes to them. Bucket access is therefore a **read** grant through an IAM policy, and the pattern is about *who can read which bucket*.

Role	Bucket Access	Permissions
Developers	`dev_logs`	Read
	`staging_logs`	Read
	`prod_logs`	Read
SRE/Platform	All buckets	Read
Security Team	`security_logs`	Read
	`audit_logs`	Read
	`compliance_logs`	Read
Auditors	`audit_logs`	Read
	`compliance_logs`	Read
Finance	`payment_logs`	Read

Pattern 2: Least Privilege by Environment

```
Development Team:
  - READ access to dev_logs
  - READ access to staging_logs
  - NO access to prod_logs

QA Team:
  - READ access to staging_logs
  - READ access to dev_logs
  - READ access to prod_logs (for troubleshooting only)

Production SRE:
  - READ access to prod_logs, error_logs, security_logs
  - READ access to all other buckets
```

What a bucket can and cannot enforce

A Grail bucket is created *"with a specified name, table type (for example, logs, events, bizevents, or spans), and retention period"* — **one** retention period, 1 day to 10 years plus a week. Access is granted with IAM policies on the bucket. None of those is a minimum/maximum retention range, a write-once or immutability switch, or an encryption choice, so do not plan a compliance control around one. Requirements beyond retention and access (immutability, an access audit trail, encryption standards) are met at the platform or account level, or outside Dynatrace — confirm them with your Dynatrace account team and your compliance owner, not with a bucket setting.

Bucket	Retention	Driven by
`dev_logs`	7–14 days	Cost
`staging_logs`	14–30 days	Cost
`default_logs`	35 days	Built-in default
`prod_logs`	90 days	Operations
`audit_logs`	365+ days	Your audit window
`payment_logs`	365+ days	PCI DSS 10.5.1
`phi_logs`	2,190–2,555 days	HIPAA documentation retention

Shortening a bucket's retention **deletes** the data older than the new period — *"Shortening the retention period on update requests will delete the data that is over the new period."*

Compliance retention — what the standards say

SOC 2

SOC 2 does not prescribe a log-retention period. In community practice, audit logs are kept for at least the 12-month window a Type 2 report covers. Typical content:
- Authentication events (login, logout, MFA)
- Authorization changes (role assignments, permission changes)
- Data access (who accessed what, when)
- Configuration changes (pipeline edits, bucket changes)
- Security events (failed auth, suspicious activity)

```
Bucket:     soc2_audit_logs
Retention:  365 days or longer (match your audit window)
Access:     IAM policy — security team + auditors only
```

PCI DSS

PCI DSS Requirement 10.5.1 requires **at least 12 months** of audit log history, with at least the most recent three months immediately available for analysis. A 90-day bucket does not meet it.

- Payment transaction logs (card numbers masked at ingest — OPMIG-08)
- Access to cardholder data environments
- Authentication to payment systems
- Database queries touching payment data

```
Bucket:     pci_payment_logs
Retention:  365 days or longer
Processing: Mask card numbers and CVV in the Processing stage (OPMIG-08)
Access:     IAM policy — payment team + security + auditors
```

HIPAA

HIPAA requires covered entities to retain required documentation for six years (45 CFR §164.316(b)(2)); in community practice that period is applied to PHI access logs too, often rounded up to seven years.

- PHI access logs (who viewed which patient records)
- Authentication to healthcare systems
- Prescription and medical record changes
- Breach detection events

```
Bucket:     hipaa_phi_logs
Retention:  2,190–2,555 days (6–7 years)
Processing: Mask patient IDs, SSNs, dates of birth; hash record numbers
(OPMIG-08)
Access:     IAM policy — healthcare IT + compliance officers only
```

GDPR

Special Considerations:

- **Right to Erasure:** Logs with EU citizen PII must be deletable on request — Grail supports record deletion via API, which must be enabled as a capability in your DPS

- **Data Minimization:** Only collect necessary personal data
- **Retention Limits:** Cannot retain personal data longer than necessary

```
Bucket:     eu_customer_logs
Retention:  90 days (balancing security and data minimization)
Processing: Mask email addresses, IP addresses; add customer_id_hash (for
erasure lookup)
Erasure:    Record deletion API, driven by your erasure-request process
Residency:  Set by the tenant's hosting region — a bucket cannot change where
data is stored
```

> **Sources:** Organize data (DT docs) — *"Creates a new user-defined bucket with a specified name, table type (for example, logs, events, bizevents, or spans), and retention period."*, *"For custom buckets, the possible retention periods range from 1 day to 10 years, with an additional week."*, *"Shortening the retention period on update requests will delete the data that is over the new period."*; Record deletion in Grail via API (DT docs) — *"This must be enabled as a capability in your Dynatrace Platform Subscription (DPS)."* PCI DSS v4.0.1 Requirement 10.5.1 and 45 CFR §164.316(b)(2) are cited from the standards themselves.

Bucket Naming Conventions

Recommended Pattern: `<environment>_<purpose>_logs`

Examples:

```
prod_app_logs
prod_error_logs
prod_security_logs
staging_app_logs
dev_debug_logs
compliance_audit_logs
compliance_soc2_logs
compliance_pci_logs
compliance_hipaa_logs
ephemeral_healthcheck_logs
```

Benefits:
- Clear ownership and purpose
- Easy to identify in DQL queries

- Simplifies access control rules
- Supports automated governance checks

Bucket Lifecycle Management

Quarterly Review Checklist:
- [ ] Review bucket usage and growth trends
- [ ] Validate retention periods still match requirements
- [ ] Check for unused or abandoned buckets
- [ ] Audit access control list changes
- [ ] Verify compliance bucket configurations
- [ ] Analyze query patterns for optimization opportunities
- [ ] Review storage costs vs. budget

Annual Compliance Audit:
- [ ] Verify all compliance buckets meet regulatory retention
- [ ] Confirm masking/redaction rules are working
- [ ] Test data erasure procedures (GDPR)
- [ ] Review audit trail completeness
- [ ] Validate encryption standards
- [ ] Document bucket governance changes

Best Practices

Routing Best Practices

Practice	Description
Specific first	Place specific routes before general ones
Test conditions	Validate matching conditions with sample data
Avoid overlap	Unless intentional multi-pipeline is needed
Document routes	Keep a map of route → pipeline → bucket
Monitor the default route	High default-route volume = missing routes

Bucket Best Practices

Practice	Description
Meaningful names	Use descriptive bucket names
Match retention to value	Critical data = longer retention
Consider query patterns	Frequently queried data in main bucket
Plan for compliance	Audit logs may need 365+ days
Review regularly	Re-evaluate bucket strategy quarterly

Cost Optimization Best Practices

Practice	Impact
Drop DEBUG/TRACE	Volume reduction equal to your DEBUG/TRACE share — measure it with the queries above
Drop health checks	Varies widely by workload — measure before you plan to it
Short retention for dev	Storage for that data falls in proportion to retention (7 of 35 days ≈ 80% less)
Extract metrics, don't store logs	Metrics persist; assign the raw logs **No storage assignment** (a Drop record processor would run before extraction)
Sample high-volume noise	Keep visibility, reduce storage

Routing Configuration Examples

Example 1: Environment-Based Routing

Route 1: Production Logs

```
Condition: k8s.namespace.name == "production" OR k8s.namespace.name == "prod"
Pipeline: production-logs
Bucket: production_logs (35 days)
```

Route 2: Development Logs

```
Condition: k8s.namespace.name == "development" OR k8s.namespace.name == "dev"
Pipeline: development-logs
Bucket: dev_logs (7 days)
```

Example 2: Severity-Based Routing

Route 1: Error Logs (high value)

```
Condition: status == "ERROR"
Pipeline: error-logs
Bucket: error_logs (90 days)
Processing: Parse stack traces, enrich with service info
```

Route 2: Debug Logs (low value)

```
Condition: loglevel == "DEBUG" OR loglevel == "TRACE"
Pipeline: debug-logs
Processing: DROP all (or route to 3-day bucket)
```

Example 3: Compliance Routing

Route 1: Audit Logs

```
Condition: matchesValue(log.source, "*audit*") OR matchesPhrase(content,
"authentication")
Pipeline: audit-logs
Bucket: audit_logs (365 days)
Processing: Mask PII, add compliance tags
```

Next Steps

Now that you understand routing and buckets, continue with:

Notebook	Focus Area
OPMIG-06	Processing, Parsing & Transformation
OPMIG-07	Metric & Event Extraction
OPMIG-08	Security, Masking & Compliance
OPMIG-09	Troubleshooting & Validation

References

- OpenPipeline Data Flow
- organize-data (DT docs)
- OpenPipeline Limits

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.